Assignment-8

Q.3.

$$I_o = \frac{500}{100} = 10A$$

leading leg will go for soft switching, thus

$$t_{deadtime\ (min.)} = \frac{2 \times C_{oss} \times V_{dc}}{I_o}$$

(for achieving soft switching)

$$= \frac{2 \times 100 \times 10^{-12} \times 200}{5}$$

$$= 8 \times 10^{-9}$$

$$t_{deadtime\ (min.)} = 8ns$$

Q.4.

$$t_{deadtime\ lagging} \geq \frac{1}{\omega_1} \sin^{-1}\left(\frac{V_{in}}{I_2}\sqrt{\frac{2C_{oss}}{L_{lag}}}\right)$$

$$\omega_1 = \frac{1}{\sqrt{2L_{lag}C}} = \frac{1}{\sqrt{2 \times 10 \times 10^{-6} \times 500 \times 10^{-12}}}$$

$$\omega_1 = 10000000$$

$$I_2 \simeq I_o = \frac{50}{10} = 5$$

$$t_{deadtime\ lagging} = \frac{1}{10^7} \sin^{-1}\left(\frac{100}{5}\sqrt{\frac{2 \times 500 \times 10^{-12}}{10 \times 10^{-6}}}\right)$$

$$= \frac{1}{10^7} \frac{\sin^{-1}(0.2)}{57.3^\circ}$$

$$= 20.13 \times 10^{-9}$$

$$t_{deadtime\ (lagging} = 20.13\ ns$$

Q.5.

$$D_{loss} = \frac{4 L_{Mag} \cdot n I_0 f_{sw}}{V_{in}}$$

$$0.05 = \frac{4 \times L_{Mag} \times 1 \times 5 \times 50000}{200}$$

$$L_{Mag} = 10\,\mu H$$

Q.6.

$$1 - D = \frac{\delta}{180}$$

$$\Rightarrow 1 - D = \frac{180}{180}$$

$$\Rightarrow D = 0$$

$$\Rightarrow V_0 = 0$$

Q.8.

$$\omega = \frac{1}{\sqrt{L \cdot 2C}} \Rightarrow$$

$$2\pi \times 100000 = \frac{1}{\sqrt{L \times 200 \times 10^{-9}}}$$

$$L = 12.67\ \mu H$$

$$\Rightarrow L_{series} = (12.67 - 5) \times 10^{-6} \quad \text{(5 → leakage give)}$$

$$\boxed{L_{series} = 7.67\ \mu H}$$

Q.10.

$$1 - D = \frac{\delta}{180}$$

$$\Rightarrow 1 - D = \frac{72}{180} = 0.4$$

$$\Rightarrow D = 0.6$$

$$\Rightarrow V_0 = n V_{in} \cdot D$$

$$\Rightarrow \frac{V_0}{V_{in}} = 1 \times 0.6$$

$$\Rightarrow \boxed{\frac{V_0}{V_{in}} = 0.6}$$